

MTH:2310 DISCRETE MATH
QUIZ 1
DUE: MONDAY FEB. 3

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: (4 points)

Your ability to express your solutions is just as important as mathematical accuracy.

You will be penalized up to 2 points for arithmetic errors.

You will be penalized up to 2 points for disorganized or hard to read solutions.

Problem 1. (12 points) *Propositions*

Determine whether the following statements are propositions. If the statement is a proposition, indicate whether it is (T)rue or (F)alse.

- a. (3 points) "The Milwaukee School of Engineering is located in Salt Lake City, Utah."

This is a proposition. It can be verified,
and is False.

- b. (3 points) "This is a true statement."

Let $p =$ "This is a true statement."

Suppose $p = T$. Statement claims that $p = T$.

The claim is true so $p = T$.

Then p is true.

Suppose $p = F$. Statement claims $p = T$.

The claim is false so $p = F$.

Then p is false.

p is true and false, so not a proposition

- c. (3 points) "If $x = 5$, then $5 + x = 10$."

This is a true proposition.
Verified algebraically

- d. (3 points) "Please write your answer for Problem 1(d) below."

Not a proposition, not a declarative statement. Just a command.

Problem 2. (12 points) Compound Propositions

Translate each statement into a compound proposition (using propositional variables and logical operators). Clearly define every variable you use.

- a. (4 points) "I am early to class or there is traffic on the interstate."

p = "I am early to class."
 q = "There is traffic on the interstate."

$$p \vee q$$

- b. (4 points) "You did not get an A on the exam and you studied for 10 hours."

p = "You did not get an A on the exam."
 q = "You studied for 10 hours."

$$p \wedge q$$

c. (4 points) "If I stub my toe, I will cry for at least 2 minutes."

p = "I stub my toe."
 q = "I cried for at least 2 minutes."
 $p \Rightarrow q$

Problem 3. (6 points) Compound Propositions

If p = "The sun is extremely hot." and q = "I will get a very serious sunburn today." translate the following compound propositions into words.

a. (3 points) $p \wedge q$

The sun is extremely hot and I will get a very serious sunburn today.

b. (3 points) $\sim q \leftrightarrow \sim p$

I will not get a very serious sunburn today only if the sun is not extremely hot.

c. (3 points) $(p \rightarrow q) \vee q$

If the sun is extremely hot, I will get a very serious sunburn today, or I will get a very serious sunburn today anyway.

Problem 4. (12 points) Truth Tables

Fill out a truth table for the following propositional expressions.

- a. (4 points)
- $(\sim p \leftrightarrow q) \oplus (q \wedge p)$

p	q	$\sim p$	$q \wedge p$	$\sim p \leftrightarrow q$	$(\sim p \leftrightarrow q) \oplus (q \wedge p)$
T	T	F	T	T	F
F	T	T	F	T	T
T	F	F	F	T	T
F	F	T	F	F	F

- b. (8 points)
- $(\sim p \leftrightarrow \sim q) \leftrightarrow (q \leftrightarrow r)$

Hint: A truth table with n distinct propositions must have 2^n rows.

p	q	r	$\sim p \leftrightarrow \sim q$	$q \leftrightarrow r$	$(\sim p \leftrightarrow \sim q) \leftrightarrow (q \leftrightarrow r)$
T	T	T	T	T	T
F	T	T	F	T	F
T	F	T	F	F	T
F	F	T	T	F	F
T	T	F	T	F	F
F	T	F	F	F	T
T	F	F	F	T	F
F	F	F	T	T	T